

User Manual

Floppy Disk Drive Emulator SFR1M44-DU26 (PCB SFRC2D.B)

Installation, Operation and Reference

TDS Floppy Emulator Firmware 1.2.0

TFE-0002-00

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The SFR1M44-DU26 hardware is commonly sold under the Gotek name. All names are the property of their respective owners and are used only to identify compatible equipment.

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General Safety Summary

Review the following precautions to avoid injury and to prevent damage to the emulator and to the equipment it is installed in. Follow also the safety information of that equipment.

Disconnect Power. Switch the host equipment off and disconnect its power cord before opening it and before connecting or disconnecting the emulator's interface cable.

Do Not Connect Under Power. The interface cable carries the emulator's +5 V supply. Connecting or disconnecting it with power applied can damage the emulator or the host.

Prevent Electrostatic Discharge. The emulator and the host circuit boards are sensitive to static discharge. Handle them only in a static-controlled environment, wearing a grounded wrist strap.

Keep Your Data. Some operations replace the contents of the emulator's internal disk. Save anything you need with DATA OUT first.

Preface

This manual describes the floppy disk drive emulator running the TDS Floppy Emulator Firmware: what it does, how to install it in any equipment that uses a 26-pin slim floppy disk drive, how to operate it, and the details of its interface, disk formats and files for reference.

Chapter 1 describes the product and what the host equipment must provide. Chapter 2 covers loading the firmware and installing the emulator; Chapter 3, operation; Chapter 4, troubleshooting; Chapter 5, specifications. The appendices give the interface signals and timing, the disk and file system formats, and the rules and formats of the files the emulator reads and writes.

Names of controls, indicators and files are shown as they appear on the emulator or the flash drive, for example DATA IN, UPDATE . UPD.

This document applies to firmware version 1.2.0 and above.

Related Manuals

The *TDS Floppy Disk Drive Emulator Instructions* (TFE-0001-01) describe the installation of the emulator in Tektronix TDS 500, 600 and 700 series oscilloscopes and the TLS 216, with the instrument-specific procedures. The project repository (github.com/JaredCabot/tds-floppy-emulator) holds the firmware releases, the change log, the design notes and the source code.

Product Description

Overview

The floppy disk drive emulator replaces a 3.5 inch floppy disk drive that connects through a 26-pin flat flex cable. To the host equipment it is a floppy disk drive with a disk inserted: the host reads, writes and formats that disk exactly as it would a real one, at 1.44 Mbyte (HD) or 720 kbyte (DD). The disk is held in the emulator's internal flash memory, so it keeps its contents without power and has no moving parts.

Two buttons exchange files between the internal disk and a USB flash drive plugged into the front of the emulator: DATA IN loads files from the flash drive onto the internal disk, and DATA OUT saves the files on the internal disk to the flash drive. A red indicator shows transfers and their results. The same USB port and buttons install firmware updates.

Features

- Internal disk of 1.44 Mbyte or 720 kbyte in flash memory, written through a power-safe journal: a power loss affects only the write in progress.
- Density follows the disk, as with a real drive: the host formats the disk at the density its drive-type line selects (jumper JE), and the emulator follows.
- DATA IN: loads the files from the flash drive's root directory in alphabetical order, as many as fit, with every sector verified; further presses load the next group.
- DATA OUT: saves every file on the internal disk, directories included, and verifies each copy; existing files on the flash drive are never overwritten, and files already there are not copied twice. Only then is the internal disk erased.
- A transfer can be cancelled with its own button. Nothing happens when a flash drive is inserted until a button is pressed.
- Flash drives formatted FAT12, FAT16, FAT32 or exFAT, with MBR or GPT partitioning.
- Firmware updates from a flash drive, protected against power loss; a watchdog; a flash memory fault is reported and remembered.

Host Requirements

The emulator works in place of a drive in equipment that provides the following. It has been verified with Tektronix TDS 784D and TDS 794D oscilloscopes; reports from other equipment are welcome in the project repository.

Table 1–1: Host requirements

Item	Requirement
Interface	26-pin slim floppy interface on a 1 mm pitch flexible flat cable, with the signal assignment of Table A–1: DISK CHANGE on pin 6, READY on pin 8
Power	+5 V on the interface cable (pins 1, 3 and 5)
Disk type	3.5 inch 1.44 Mbyte (2HD) or 720 kbyte (2DD), PC format: MFM, 300 rpm, 512-byte sectors, 18 or 9 sectors per track, 80 cylinders, 2 heads
File transfers	A FAT12 (PC) file system on the disk, as the host creates when it formats. A host that uses its own file system can still use the emulator as a drive, but not DATA IN and DATA OUT
Mounting	Space for the emulator in place of the drive; see <i>Connect and Mount the Emulator</i>

NOTE. Other disk formats (for example 256-byte sectors or other track counts) are not supported.

Installation

Load the Firmware

Load the firmware (once, if the emulator still has its factory firmware), set the jumpers for the host, then connect and mount the emulator in place of the drive and verify it.

An emulator supplied with the TDS Floppy Emulator Firmware already installed does not require this procedure. An emulator supplied with its factory firmware must be loaded once, before it is installed, through the USB bootloader built into its microcontroller. Later firmware versions are installed from a USB flash drive (see *Updating the Firmware*).

Table 2-1: Load the firmware: requirements

Item	Requirement
Equipment required	PC running Windows; USB-A to USB-A cable; one jumper link (2.54 mm)
Software required	Artery ISP Programmer (Artery Technology), including its DFU driver installer (archived in the project repository)
Firmware file	<code>tdsfloppy_install.hex</code> from the latest release (bootloader and firmware in one file)
Prerequisites	The emulator is not installed in the host and no USB flash drive is connected

A USB-A to USB-A cable is not common, but is easily made from two USB-A plugs or two cut USB cables. It is wired straight through, pin to pin:



CAUTION. *The PC powers the emulator through this cable. Use it only while the emulator is disconnected from any host, and never between two PCs or other USB hosts: both ends would supply +5 V.*

Table 2-2: USB-A to USB-A cable wiring

Pin	Signal	Usual wire colour	Connects to
1	VBUS (+5 V)	Red	Pin 1 of the other plug
2	D-	White	Pin 2 of the other plug
3	D+	Green	Pin 3 of the other plug
4	GND	Black	Pin 4 of the other plug
Shell	Shield	Braid or drain wire	Shell of the other plug (optional)



CAUTION. *The factory firmware is erased permanently by this procedure. The factory firmware cannot be read out of the emulator and is not required for any later step.*

1. *Select the bootloader:* Fit a jumper link across pins 1 and 2 of header **J3** on the emulator circuit board (pin 1 is BOOT0, pin 2 is +3.3 V; see Figure 2-1).
2. *Install the DFU driver:* Run `Artery_DFU_DriverInstall.exe` from the `Artery_DFU_DriverInstall` folder of the Artery ISP Programmer package, and follow its prompts.
3. *Connect the emulator:* Connect the emulator USB port to the PC with the USB-A to USB-A cable. The emulator is powered from the cable and is detected by the PC as a DFU device.
4. *Select the device:* Start the Artery ISP Programmer. Set the port type to **USB DFU**, select the emulator in the device list, and press **Next** until the operation page appears.
5. *Remove the access protection:* Select **Enable/Disable Protection**; then select **DISABLE** and **Access Protection**, press **Next**, and confirm with **Yes**. The flash memory is erased. When the erase completes, press **Back**.
6. *Program the firmware:* Select **Download to device**, press **Add**, select `tdsfloppy_install.hex`, and press **Next**. The file is programmed and verified. Press **Ok**; then press **Close**.
7. *Restore normal start-up:* Disconnect the USB cable and remove the jumper link from header **J3**.

NOTE. *Use `tdsfloppy_install.hex`, not `tdsfloppy.hex`. The latter holds the firmware without its bootloader and does not start when programmed on its own.*

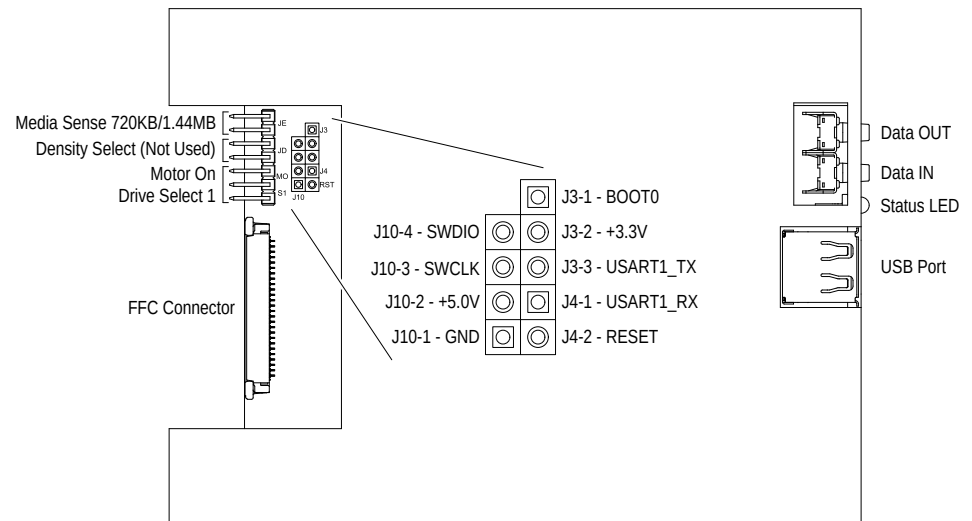


Figure 2–1: Emulator jumpers, headers and controls

Set the Jumpers

The jumper block at the rear of the emulator, next to the 26-pin connector, selects which interface line selects the drive (S1 or MO) and sets the drive type the host sees (JE); JD is not needed (see Figure 2–1). None of them is read by the firmware: they connect interface lines directly (Table A–1).

Table 2–3: Jumper positions

Position	Pins	Function	Setting
S1	Three-pin header, pins nearest the connector	The drive responds to DRIVE SELECT (pin 4)	Fitted for most hosts
MO	Three-pin header, other two pins	The drive responds to MOTOR ON (pin 10) instead	Only for hosts that select the drive by motor on
JE	Two-pin header	HD OUT (pin 9) low: the host sees a 720 kbyte (DD) drive	Fitted for 720 kbyte disks
JD	Two-pin header	Holds pin 11 low; not used by the emulator (Table A–1)	Not fitted

S1 and MO share the three-pin header: fit exactly one. Fit JE only for 720 kbyte disks (*Disk Density*, Chapter 3).

Connect and Mount the Emulator



CAUTION. Switch the host off and disconnect its power cord first. The interface cable carries the emulator's supply.

1. *Remove the drive:* Remove the floppy disk drive from the host as its service documentation describes. Keep its screws and any mounting bracket.
2. *Connect the cable:* Insert the host's flexible cable into the emulator's 26-pin connector as far as possible, then gently press the connector cam into place to secure it. When inserted correctly, the side ears of the cam bottom out on the connector body. Verify that the cable and its contact fingers are straight.
3. *Mount the emulator:* Fit the emulator in place of the drive with the drive's screws, so that its front panel, with the USB port and the two buttons, is reachable. The emulator is narrower than many drives; some hosts need an adapter bracket (the project repository has a 3D-printable bracket for the Tektronix TDS oscilloscopes).
4. *Close the host:* Reassemble the host and reconnect its power.

Verify the Installation

1. *Switch on:* Switch the host on. The red indicator stays off (a fast double blink means no valid firmware: load the firmware again).
2. *Read the disk:* Open the host's file or disk menu. A new emulator holds an empty disk; the host lists no files and no errors.
3. *Write the disk:* Save any file to the disk and list the directory. The file appears with its size.

If the host reports no drive or no disk, check the cable, the jumper at S1 (or MO) and Chapter 4.

Operation

Controls and Indicators

The front panel carries a USB port, two buttons and the indicators. The DATA OUT button is the one next to the USB port; DATA IN is the other.

Table 3-1: Controls

Control	Function
DATA IN button	Press and release: replace the internal disk with the next group of files from the flash drive.
DATA OUT button	Press and release: save every file on the internal disk to the flash drive, verify them, then erase the internal disk.
The same button again	While the red indicator flashes: cancel the transfer (see <i>Cancelling a Transfer</i>).
Both buttons	Hold 3 seconds until the red indicator lights steadily, then release: update the firmware, or, with no update file on the flash drive, write a status report.
Red indicator	Transfers and their results (see <i>Indicator Patterns</i>).
Green indicator (if fitted)	Lit while the host selects the drive. It is wired to the select line, not controlled by the firmware; some units have none.

A button acts when it is released. Inserting a flash drive does nothing by itself. If the host is using the drive when a button is pressed, the emulator waits (the red indicator flashing) until the host has been idle for 1.5 seconds, then starts; if the host is still busy after 15 seconds, the transfer is refused with the error pattern.

The Internal Disk

The internal disk behaves as a floppy disk that never leaves the drive. The host reads, writes and formats it; its contents remain when the power is off. Each track the host writes is stored in flash memory shortly after the write ends, through a journal: if power is lost at any moment, only the write in progress can be affected, never the rest of the disk. A new emulator holds an empty disk.

During a transfer the emulator removes the disk from the host, as if it had been ejected, and inserts it again afterwards; the host sees a disk change.

Disk Density

Like a real floppy disk, the internal disk is either 1.44 Mbyte (HD) or 720 kbyte (DD). Jumper JE tells the host which kind of disk to expect: without JE, 1.44 Mbyte; with JE fitted, 720 kbyte.

Changing the jumper alone does not change the disk. The disk takes on a new density when the host formats it, just as a real floppy disk does when it is formatted in a drive: the emulator recognises the density the host is writing and follows it.

1. *Save the files:* Save anything needed from the disk with DATA OUT. Formatting erases the disk.
2. *Set the jumper:* With the host switched off, fit JE for 720 kbyte or remove it for 1.44 Mbyte.
3. *Format the disk:* Switch the host on and format the disk with the host's format command. The disk is now of the new density.

The density is kept through power-off, firmware updates, DATA IN and DATA OUT. A host that reads the drive type only at power-on must be switched off and on after the jumper is changed. If the jumper and the disk disagree, the host reads the disk as unformatted until it formats it, as with a real disk of the wrong type.

USB Flash Drive Requirements

Table 3–2: Flash drive requirements

Item	Requirement
File system	FAT12, FAT16, FAT32 or exFAT (NTFS is not supported)
Partitioning	MBR, GPT, or none
Sector size	512 bytes (all common flash drives)
Capacity	Up to 2 Tbyte
Files for DATA IN	In the root directory

A flash drive that cannot be read (NTFS, unformatted, or another sector size) is reported with three slow blinks; nothing is changed.

Loading Files from a USB Flash Drive



CAUTION. *DATA IN replaces the entire contents of the internal disk. Save anything needed with DATA OUT first.*

1. *Prepare the flash drive:* Copy the files to the root directory of the flash drive.
2. *Load the files:* Insert the flash drive and press and release DATA IN. The red indicator flashes while the files are copied and verified.
3. *Use the files:* When the indicator goes out, the host sees a disk holding the loaded files.
4. *Load more:* If the flash drive holds more than fits on the disk, press DATA IN again for the next group; after the last group, the first follows.

Files are loaded in alphabetical order, regardless of case, as many as fit on the disk (1.44 Mbyte, or 720 kbyte for a DD disk). Directories, hidden and system files, UPDATE . UPD, the status report EMUSTAT . TXT and files larger than the disk are skipped. Long file names are given short 8.3 names, as Windows does (Appendix C). Every sector is read back and compared as it is written. If a load fails or is cancelled, the next press loads the same group again.

Saving Files to a USB Flash Drive

1. *Insert the flash drive:* Insert a flash drive with free space.
2. *Save the files:* Press and release DATA OUT. The red indicator flashes while the files are copied and verified.
3. *Check the result:* When the indicator goes out without an error pattern, every file is on the flash drive and the internal disk is empty, ready for new data.

DATA OUT copies every file on the internal disk, keeping its date and time, and recreates the directories on the flash drive (up to seven levels below the root; an existing directory of the same name is used; directories nested more deeply are skipped, the rest is saved, and the error pattern is shown with the disk kept). Files on the flash drive are never overwritten: a file whose name exists is saved as NAME_1.EXT, NAME_2.EXT and so on, and a file already there with the same name and the same contents is not copied again. Each copy is read back and compared. The internal disk is erased only when every file is safely on the flash drive; after any error it is left unchanged, and no partial copy remains.

Cancelling a Transfer

Press the button that started a transfer again while the red indicator flashes; the transfer stops within a fraction of a second and the indicator gives one long blink. The other button has no effect during a transfer.

- DATA OUT: any partly copied file is deleted; files already copied remain on the flash drive; the internal disk is not erased. A later DATA OUT does not copy the files already on the flash drive again.
- DATA IN: the internal disk keeps the files loaded so far, and the next press loads the same group again. The previous contents cannot be restored: DATA IN erases the disk when it starts.
- A firmware update cannot be cancelled.

Formatting the Internal Disk

Format the disk with the host's own format command, as a floppy disk. The format sets the density (see *Disk Density*). An empty disk can also be obtained with DATA OUT, which erases the disk after saving its files.

Updating the Firmware

A firmware update is a file named UPDATE . UPD, published with each release. The update is checked completely before anything is changed: a damaged file or a file for other hardware is refused. The internal disk and its files are kept.

1. *Prepare the flash drive:* Copy UPDATE . UPD to the root directory of the flash drive and insert it in the emulator.
2. *Arm the update:* Press and hold both buttons. After 3 seconds the red indicator lights steadily.
3. *Start the update:* Release both buttons. The red indicator flashes while the file is checked and copied; the emulator then restarts and installs the firmware, taking about 5 seconds in all.
4. *Check the result:* Check that the emulator resumes normal operation. Six quick blinks mean the file was refused, and the installed firmware is unchanged. With no UPDATE . UPD on the flash drive, the emulator writes a status report instead (see *Status Report*).

NOTE. A power loss during the update does not damage the emulator: either the installed firmware is kept, or the installation completes at the next power-on. Going back to firmware 1.0.0, which does not know 720 kbyte disks, makes a 720 kbyte disk read as unformatted until it is updated again or formatted at 1.44 Mbyte.

Status Report

Holding both buttons with no `UPDATE .UPD` on the flash drive writes a status report to it instead, a text file named `EMUSTAT .TXT` in the root directory, replaced each time. The red indicator flashes while it is written and then goes out. The internal disk is not changed, although the host sees a disk change. The report lists:

- the firmware version and build ID, the board, the emulator's unique serial number, and its clock source (the crystal, or the internal oscillator if the crystal did not start);
- the internal disk: density, number of files and directories, bytes used and free;
- health: whether a flash memory fault is recorded, internal counters, the cause of the last restart, and the time since power-on;
- the flash drive's file system and capacity, and the result of the last `DATA IN` or `DATA OUT`.

Include the report when asking for help with an emulator.

Indicator Patterns

Table 3–3: Red indicator patterns

Pattern	Meaning
Off	Idle
Flashing, 4 times per second	Transfer in progress; do not remove the flash drive
On steadily	Both buttons held 3 seconds; release to update the firmware
2 quick blinks	Nothing to load (no suitable files); internal disk unchanged
3 slow blinks	Flash drive format not supported; reformat as exFAT or FAT32; internal disk unchanged
1 long blink (1 s)	Transfer cancelled
6 quick blinks	Error: no flash drive, USB error, verification failure, update refused, or drive in use for more than 15 seconds; internal disk unchanged
Fast blinking, continuous	Internal flash memory failing: a track could not be stored (kept after power-off until the disk is rebuilt; save the files with DATA OUT), or the power-on self-test failed (no disk is presented)
Fast double blink, repeating	No valid firmware installed; load the firmware

Troubleshooting

Table 4–1: Troubleshooting

Symptom	Possible cause	Action
Host reports no drive or a drive error; green indicator (if fitted) never lights	Cable not seated, or the select jumper at the wrong position	Reseat the cable in the connector cam; fit the jumper at S1 (or MO for hosts that select by motor on)
Host reads the disk as unformatted	Jumper JE does not match the disk's density, or the disk was never formatted by this host	Set JE for the wanted density and format the disk in the host (Disk Density)
Directory shows unexpected files	A DATA IN replaced the disk	Load the required group with DATA IN, or save with DATA OUT
A file is missing after DATA IN	Larger than the disk, hidden, in a directory, or in a later group	Press DATA IN again for the next group; move the file to the root directory
2 quick blinks after DATA IN	No suitable files in the root directory	Copy the files to the root directory of the flash drive
EMUSTAT . TXT appears instead of an update	No UPDATE . UPD in the root directory of the flash drive	Copy UPDATE . UPD to the root directory and hold both buttons again
3 slow blinks	Flash drive is NTFS, unformatted, or not 512-byte sectors	Reformat the flash drive as exFAT or FAT32
6 quick blinks	No flash drive, USB error, verification failure, update refused, or the host kept using the drive	Reinsert the flash drive, wait for the host to finish, and press again; try another flash drive
Copies named NAME_1.EXT on the flash drive	A different file of that name already existed	Normal operation; existing files are never overwritten
Fast double blink at power-on	No valid firmware	Load the firmware (Chapter 2)
Fast continuous blinking	Internal flash memory failing	Save the files with DATA OUT; if it recurs, replace the emulator

Specifications

Table 5–1: Emulated drive

Characteristic	Description
Drive	3.5 inch floppy disk drive, 300 rpm, MFM
1.44 Mbyte (HD)	500 kbit/s; 80 cylinders, 2 heads, 18 sectors of 512 bytes per track (HD OUT high: JE not fitted)
720 kbyte (DD)	250 kbit/s; 80 cylinders, 2 heads, 9 sectors of 512 bytes per track (HD OUT low: JE fitted)
Head travel	Cylinders 0 to 80
Ready	READY 480 ms after drive select, as a drive after motor spin-up
Write protection	None (the disk is always writable)
Write storage	Starts 0.1 s after each write ends, through a power-safe journal; the head never waits for flash memory

Table 5–2: Storage, USB and power

Characteristic	Description
Internal disk	1.44 Mbyte or 720 kbyte in SPI flash memory, retained without power
USB	USB 2.0 full-speed host, mass-storage class, type A socket
Flash drives	FAT12, FAT16, FAT32, exFAT; MBR, GPT or unpartitioned; 512-byte sectors; up to 2 Tbyte
Transfer rate	DATA IN approximately 95 kbyte/s (a full 1.44 Mbyte disk in about 15 s); DATA OUT approximately 150 kbyte/s, each copy read back and compared
Firmware update	UPDATE . UPD; header and CRC checked before anything changes; about 5 s; protected against power loss
Power	+5 V from the interface cable; from the USB cable while the firmware is loaded
Interface	26-pin slim floppy interface, 1 mm pitch flexible cable, cam-lock connector

Appendix A: Interface Reference

Signals are active low unless stated. Outputs (from the emulator) are driven only while the drive is selected, except HD OUT, which is set by jumper JE alone; inputs are ignored while the drive is not selected.

Table A-1: Interface connector pin assignments

Pin	Signal	Direction	Description
1, 3, 5	+5 V	Power	Emulator supply
2	INDEX	Out	One pulse per revolution
4	DRIVE SELECT	In	Selects the drive (jumper S1)
6	DISK CHANGE	Out	Disk changed since the last step
7, 13	(none)		Not connected
8	READY	Out	Drive ready
9	HD OUT	Out	Media sense. High: HD (JE open); low: DD (JE fitted)
10	MOTOR ON	In	Selects instead of pin 4 (jumper MO)
11	DENSITY SELECT	In	Not used; held low while JD is fitted
12	DIRECTION	In	Low: step toward the centre
14	STEP	In	Head step, on the rising edge
16	WRITE DATA	In	Write flux transitions
18	WRITE GATE	In	Write enable
20	TRACK 00	Out	Head at cylinder 0
22	WRITE PROTECT	Out	Never asserted
24	READ DATA	Out	Read flux transitions
26	SIDE ONE SELECT	In	Low: head 1
15-25 odd	0 V	Ground	Pins 15, 17, 19, 21, 23, 25

Table A-2: Signal timing

Signal	Behaviour
READY	Asserted 480 ms after the drive is selected; dropped when it is deselected. INDEX and READ DATA are silent until READY
INDEX	Pulse of about 3 ms every 200 ms (300 rpm)
STEP	Acts on the rising edge; step rates of 3 ms and slower
Seek complete	INDEX and READ DATA are suppressed for 17 ms after the last step, as while a real head settles
READ DATA	Pulses of about 480 ns at the connector; cell 1 μ s (HD) or 2 μ s (DD)
DISK CHANGE	Asserted at power-on and after each transfer (the disk was out); cleared by a step while selected
TRACK 00	Immediate on reaching cylinder 0
Writes	Decoded as they arrive; the track is stored in flash memory 0.1 s after the write gate goes inactive

Appendix B: Disk Formats

The emulator generates each track in the standard PC (IBM System 34) MFM layout. It recognises the density of a disk the host formats from the data rate of the host's writes: only a write spanning at least three quarters of a revolution at the other rate (a format) changes it.

Table B-1: Track layout

Item	1.44 Mbyte (HD)	720 kbyte (DD)
Data rate	500 kbit/s	250 kbit/s
Bytes per revolution	12,500	6,250
Sectors per track, numbered	18, from 1	9, from 1
Sector size	512 bytes	512 bytes
Gap 4a, sync, index mark, gap 1	146 bytes	146 bytes
ID field + gap 2 + data field	22 + 22 + 530 bytes	22 + 22 + 530 bytes
Gap 3	84 bytes	80 bytes
Cylinders, heads	81 (0 to 80), 2	81 (0 to 80), 2

DATA IN builds, and DATA OUT reads, FAT12 volumes with the following parameters. DATA OUT also reads any other valid FAT12 volume of 512-byte sectors with 1 or 2 sectors per cluster.

Table B-2: FAT12 volumes built by DATA IN

Parameter	1.44 Mbyte	720 kbyte
Sectors	2,880	1,440
Sectors per cluster	1	2
FATs x sectors	2 x 9	2 x 3
Root directory entries	224	112
Media byte	F0	F9
First data sector	33	14
Data clusters	2,847	713
Largest file	1,457,664 bytes	730,112 bytes

Appendix C: Transfer and Update Files

File Names

The internal disk holds 8.3 names. DATA IN gives a flash drive file with a longer or non-8.3 name the short name Windows would give it: up to six characters of the name, a tilde and a number, and up to three characters of the extension, in capitals (for example `Scope capture 1.bmp` becomes `SCOPEC~1.BMP`). Characters that are not allowed in short names become underscores.

DATA OUT writes each file under its 8.3 name. If a file of that name exists with different contents, the copy is named `NAME_1.EXT`, then `NAME_2.EXT`, up to `NAME_999.EXT`; a file with the same name and contents is already saved and is not copied. Directories are merged into existing ones of the same name.

Update File Format

An update file consists of a 48-byte header followed by the firmware image. All values are little-endian. The emulator checks the header, the board name, the length and both CRCs, and copies the image to a staging area and verifies it, before its bootloader installs it at the next start.

Table C-1: UPDATE.UPD header

Offset	Size	Field	Value
0	8	Magic	TDSFLUPD
8	8	Board	SFRC2D . B (other boards are refused)
16	4	Format	1
20	4	Version	Firmware version, 0x00MMmmpp (information only)
24	4	Length	Image bytes, a multiple of 4
28	4	Image CRC	CRC-32 (IEEE 802.3) of the image: the build ID
32	12	Reserved	0
44	4	Header CRC	CRC-32 of bytes 0 to 43